



O(Alpha)

Quant in your pocket.

Software-engineered strategy validation, paper execution, and audit-ready portfolio state.

- FAIL-CLOSED GATES
- DB AUDIT TRAIL
- WORKER LIFECYCLE
- CI + TESTS
- PAPER ONLY

ACCESS & ONBOARDING

01

Guided paper setup

Login, choose risk profile, run a catalog backtest, then accept the result before the dashboard unlocks.

VALIDATION HARNESS

02

Metrics are evidence

Strategy results come from committed reports with DSR, PBO, cost stress, OOS trades, and benchmark checks.

PAPER EXECUTION

03

No hidden live orders

The worker writes simulated fills, positions, cash ledger, snapshots, trades, and alerts for inspection.

LOW-RISK PASS

2 entries

h63 LGBM + proxy pass both windows

PBO GATE

.067/.077

low-risk catalog PBO across windows

BACKEND TESTS

passed

'go test ./...' verified locally

FRONTEND

typecheck

'npm run typecheck' verified locally

PAPER-AGENT COMMAND CENTER

PAPER ONLY

One active run per user

The worker evaluates daily bars, turns target weights into idempotent paper fills, and persists positions, cash ledger, snapshots, trades, and alerts.

\$100,000

DEFAULT PAPER CASH

RISK

LOW

STOP LOSS

2.5%

REBALANCE

63 bars

MOTIVATION

The problem

Retail traders see charts and signals, but rarely see whether a strategy survived costs, out-of-sample checks, overfitting tests, and safe execution constraints.

Our solution

A paper-only platform where strategy research, user onboarding, paper execution, portfolio state, and audit logs are connected through one backend.

Reviewer fix

This poster now foregrounds software engineering: architecture, tests, lifecycle safety, database design, and artifact-backed validation.

FEATURE GROUPS

ACCESS

Login, risk profile, backtest acceptance gate

RESEARCH

Harness reports, DSR/PBO, cost stress, no-lookahead

EXECUTION

Paper fills, positions, cash ledger, snapshots

OBSERVE

Dashboard state, trades, alerts, exports

SAFETY

One active agent, guarded settings, fail-closed artifacts

TESTING & EVIDENCE

GO TESTS

54 backend test files; 'go test ./...' passed

TYPECHECK

'npm run typecheck' passed for the frontend

RESEARCH

Metrics cite committed 'reports/batches' artifacts

PLANNED

Add integration + UI smoke tests before M3

SYSTEM ARCHITECTURE - RESEARCH TO PAPER EXECUTION



Research

cmd/alpha-research



Reports

JSON + MD artifacts



Catalog

risk-bucket strategy



Agent

daily worker loop



Router

idempotent fills



Postgres

ledger + snapshots



Dashboard

state + alerts



Built by Zhao Shizhen & Tan Jia Jun
Team 6645 / O(Alpha) / Milestone 2 refresh

Paper-only research system. No live brokerage orders.
A1 portrait 594 x 841 mm